

Business Summary Report: Predictive Insights for Collections Strategy

1. Summary of Predictive Insights

The predictive analysis indicates that customers with repeated missed payments, high credit utilization, low credit scores, and high debt-to-income ratios have the highest probability of becoming delinquent. Payment history was the strongest predictor, followed by credit utilization and debt burden. Customers with unstable repayment patterns should therefore be prioritized for early intervention. These findings can help the Collections team proactively identify customers before accounts become seriously overdue, improving recovery rates while maintaining positive customer relationships.

Top 3 Risk Factors

- Multiple missed payments
- High credit utilization
- High debt-to-income ratio

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Frequent missed payments	Low credit score & high debt-to-income ratio	Missed Payments, Credit Utilization, Early Intervention	High probability of delinquency, leading to increased losses

2. Recommendation Framework

Restated Insight: Customers with two or more missed payments and high credit utilization are significantly more likely to become delinquent.

Proposed Recommendation: Introduce an automated early intervention program using SMS, email, and phone reminders for high-risk customers.

Specific: Contact customers identified as high risk.

Measurable: Reduce 30-day delinquency by 10% within six months.

Actionable: Use existing CRM and collections systems.

Relevant: Supports Geldium's collections strategy and customer retention.

Time-bound: Launch within one month and review after six months.

Justification and Business Rationale: Early communication is cost-effective, scalable, and helps customers resolve payment issues before accounts become seriously overdue. This approach reduces financial losses while improving customer experience.

3. Ethical and Responsible AI Considerations

The predictive model should be monitored regularly to ensure that it performs fairly across different customer groups. Historical data may contain bias related to income or location, so fairness testing

and balanced training data should be used to reduce unfair outcomes. Logistic Regression was selected because it is transparent and allows employees to explain predictions using understandable factors such as missed payments, credit score, and credit utilization. Customer data should remain confidential, decisions should include human oversight where appropriate, and the model should comply with responsible AI principles, including transparency, accountability, fairness, and privacy.